Student Name: _____

Student ID# _____

	Check which tutorial for pick-up
T02 Monday mornings in BSB	
T03 Wednesday afternoons in ITB	
T01 Friday mornings in IWC	

CHEMICAL ENGINEERING 2O04
Midterm #1

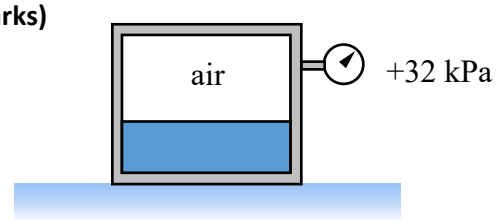
Duration 120 minutes (7:00pm – 9:00pm) **D.R. Latulippe**
McMaster University Term Test **Tuesday February 28th, 2023**

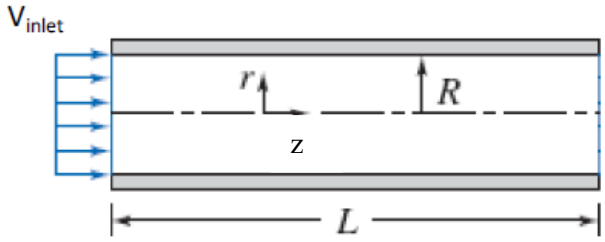
SPECIAL INSTRUCTIONS

1. The test includes 6 questions and 1 bonus question; ensure your copy of the test is complete.
2. Marks for each problem are given (**Total of 58 marks)**. Use this as a time-management guide; for example: $2\ marks\left(\frac{120\ minutes}{59\ marks}\right) = 4.1\ minutes$
3. The following materials are permitted during this examination:
   a) The Fox and McDonald's Introduction to Fluid Mechanics textbook (any edition and any version)
   b) Any printed notes or hand-written notes
   c) A tablet or computer in "airplane mode" with PDF reader to review course material. **<u>Any student caught with a device not in airplane mode and/or with other applications opened will automatically receive a grade of 0 on the test.</u>**
   d) The McMaster Standard Calculator (Casio FX-991MS/MS+)
4. **The following materials are NOT permitted during this examination:**
   a) The internet.
   b) Any software to help with calculations (e.g. Excel, MATLAB)
   c) Any applications to communicate with anyone.
5. Only tests written in pen will be considered if re-marking is requested (addition errors excepted).
6. On pages 12 & 13, there are some useful tables.

QUESTION 1 (9 MARKS)

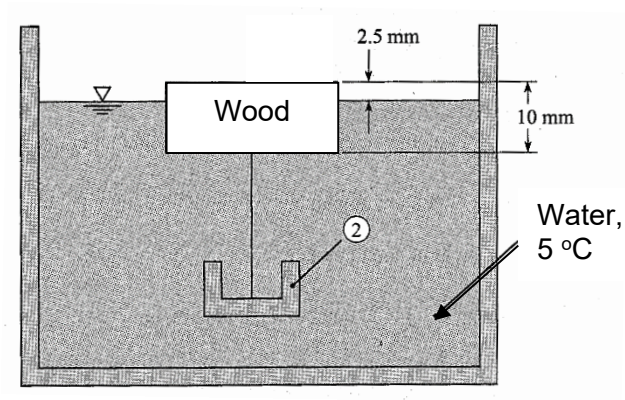
Answer the following short questions in the space provided:

a.	The kinematic viscosity (ν) of a fluid is also sometimes referred to as 'momentum diffusivity' – write Newton's law of viscosity in the correct form to show why this is. **(1 mark)**	
b.	How does the surface tension of a liquid change with increasing temperature? **(1 mark)**	
c.	A pressure gauge attached to a rigid tank displays a reading of +32 kPa as shown in the accompanying figure. Given that the tank is located at a construction site in Hamilton, Ontario (elevation ~ 100 m), determine the absolute pressure (kPa) inside the tank. **(2 marks)**  	
d.	Prove that the equation below for a falling sphere viscometer (also given on slide 9 of '04 Instrumentation' notes) is 'dimensionally consistent': **(2 marks)** $$\mu = \frac{2R^2(\rho_s - \rho)g}{9V_T}$$ where ρs and ρ are the densities of the sphere and fluid R is the radius of the sphere g is the gravitational constant V_T is the terminal velocity of the sphere	
e.	Briefly explain what a rotameter is used for and the principle by which it operates. **(2 marks)**	

f.	Water (at 4 °C) flows steadily through a pipe of length L and radius R = 7.5 mm.  If the axial velocity profile inside the pipe is correctly described by the expression $v_z(r) = V_{max}\left(1 - \left(\frac{r}{R}\right)^2\right)$ and V_{max} = 0.03 m/s, calculate the required uniform velocity at the inlet (V_{inlet}). **(1 mark)**	

QUESTION 2 (12 MARKS)

A metal part (identified by ② on the accompanying figure) is hanging by a thin cord from a floating rectangular wooden block (white pine, SG = 0.43) in a cold (T = 5 °C) liquid water bath. The total volume of the metal part is 6600 mm^3 and the dimensions of the wood block are 50 × 50 × 10 mm.



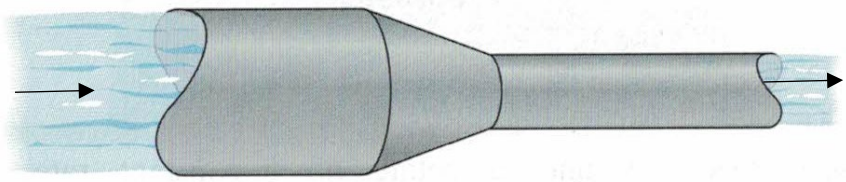
a) Draw two separate free body diagrams: one for the wood block and one for the metal part.
(4 marks)

b) Determine the magnitude of the tension force (N) of the cord that must be present to keep the wooden block in the position shown in the diagram.
(4 marks)

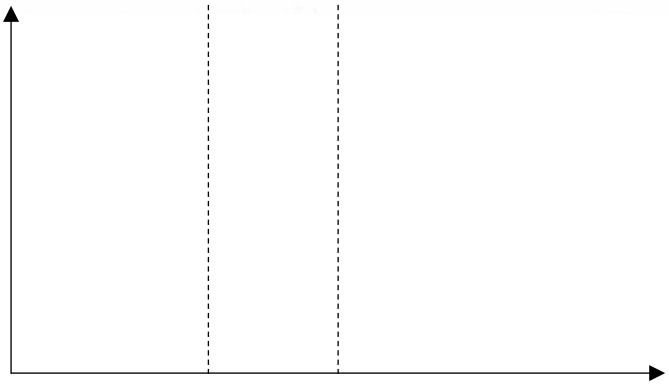
c) Using your answer from part b), determine the mass (g) of the metal part.
(4 marks)

QUESTION 3 (9 MARKS)

As shown in the accompanying figure, a liquid is flowing through a horizontal pipe with a constriction fitting that connects the larger diameter section on the left to the smaller diameter section on the right.



a. Sketch (no calculations needed) on the given graph the average velocity profile of the fluid as it flows first through the larger diameter section, then the constriction, and then the smaller diameter section. Be sure to consider the magnitude of the average velocities in developing the sketch.
(3 marks)

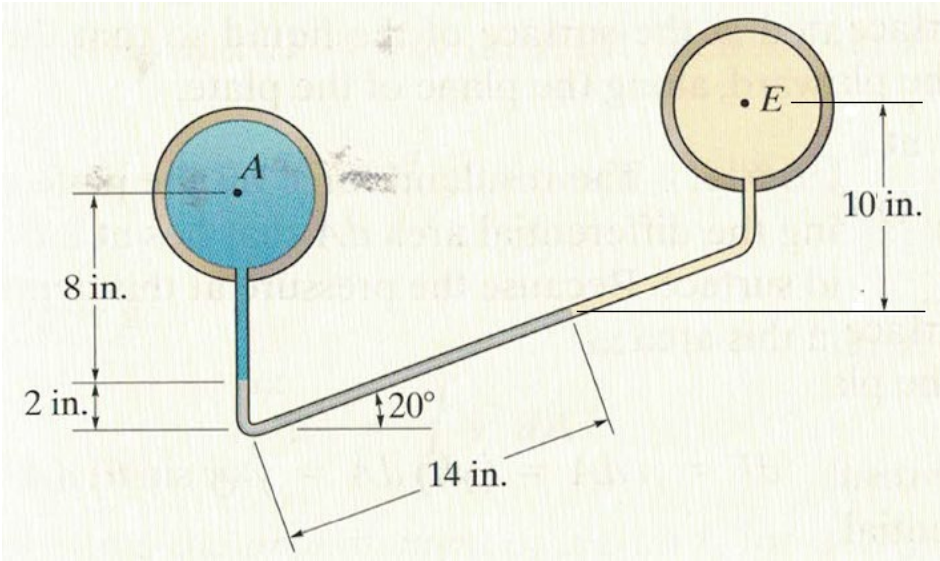


b. Give a brief explanation for why the pressure in the smaller diameter section of the pipe must be lower than that in the larger diameter section.
(2 marks)

c. As the pressure decreases due to the flow of liquid into the smaller section, there is a possibility for the liquid to spontaneously boil (a process referred to as cavitation). Determine the absolute pressure (kPa) at which this would occur if the liquid flowing through the horizontal pipe is ethanol at 35 ^{o}C.
(4 marks)

QUESTION 4 (7 MARKS)

As shown in the accompanying figure, an inclined-tube manometer filled with mercury is used to measure the pressure difference between point A (at centerline of pipe through which water at 20 °C is flowing) and point E (at centerline of pipe through which glycerine at 20 °C is flowing).



a. Calculate the pressure difference (kPa) between points A and E.

HINT: 1 in = 2.54 cm

(5 marks)

b. Now consider the scenario where $P_A = P_E$. Write a suitable equation that could be used to solve for the equilibrium position of mercury in the manometer when $P_A = P_E$.

(2 marks)

QUESTION 5 (10 MARKS)

As a junior engineer in the process automations team, you have been tasked with sourcing a suitable digital differential pressure gauge with an output signal that could be attached to an orifice meter (similar to what is shown in the accompanying figure) to measure the volumetric flow rate of kerosene (viscosity = 0.002 Pa·s).

a. Calculate the expected differential pressure (kPa) across the orifice flow meter (internal 'throat' diameter = 3.5 cm) for the current production target of 57.5 L/min of kerosene flowing through a horizontal pipe with an internal diameter of 5 cm.

(5 marks)

b. After completing the calculation described in part a), you were informed by the engineering manager that you will need to ***source a suitable digital differential pressure gauge to measure any kerosene flow rate in the range of 57.5 to 575 L/min*** (so that can be used for current and future production targets) at the exact same process conditions given in part a) (therefore the internal 'throat' diameter = 3.5 cm, pipe internal diameter = 5 cm).

Your preferred instrumentation supplier informs you that you can choose from the following four gauges with different ranges but the same 1% percentage errors:

	Pressure Range
XP10	0 to 10 kPa (error = 0.1 kPa)
XP100	0 to 100 kPa (error = 1 kPa)
XP1000	0 to 1000 kPa (error = 10 kPa)
XP10000	0 to 10000 kPa (error = 100 kPa)

Provide a suitable justification for which one of the four gauges listed above should be installed at the plant.

(5 marks)

QUESTION 6 (11 MARKS)

As shown in the accompanying figure, a water pitcher is being filled from a large jug (whose internal geometry is perfectly cylindrical) via a spout at the bottom of the jug. When the button on the spout is pressed, the velocity of water through the spout is given by $\sqrt{2gh_{jug}}$ where h_{jug} is the height of water in the jug.

a. Starting from a general <u>mass balance equation for the water in the large jug</u>, derive the appropriate equation for the change in water level (i.e. dh_{jug}/dt) in terms of the following

g	gravitational constant
D_{spout}	Internal diameter of the spout
D_{jug}	Internal diameter of the jug

(4 marks)

b. Solve the differential equation from part a) to get the correct expression for water level in the jug as a function of time (i.e. $h_{jug}(t)$) using the additional term:

$h_{jug,0}$ Initial water height

(4 marks)

c. Determine the time (in mins) that it would take to drain the large jug from an initial water height of 40 cm down to a final water height of 4 cm given the following additional specifications:

D_{spout} = 0.8 cm

D_{jug} = 30 cm

(3 marks)

BONUS QUESTION (4 MARKS)

Later in the course, we will study the topic of drag forces on objects and apply that concept to various systems.

One system that we have already studied is the falling sphere viscometer. Given that the appropriate equation for the drag force (F_D) that acts on the sphere (with density ρ_S and radius R) as it passes slowly through a viscous fluid (with density ρ and viscosity μ) at a terminal velocity V_T is given by $F_D = 6\pi\mu R V_T$, starting from Newton's 2nd Law (i.e. $\Sigma F = ma$) derive the following equation that was given on Slide 9 of the '04 Instrumentation' lecture notes:

$$\mu = \frac{2R^2(\rho_s - \rho)g}{9V_T}$$

B.4 Antoine Equation Constants

The Antoine equation

$$\log_{10} P^{sat}(\text{mmHg}) = A - \frac{B}{T(^\circ\text{C}) + C}$$

is a useful equation for modeling saturation pressures of liquids and solids. The constants should not be used outside the indicated temperature range.

Table B.4 Antoine Equation Constants for Selected Compounds

Compound	Formula	Range, °C	*A*	*B*	*C*
Acetaldehyde	CH_3CHO	−45 to +70	8.0055	1600	291.8
Acetic acid	CH_3COOH		7.38782	1533.313	222.309
Acetic anhydride	$C_4H_6O_3$		7.14948	1444.718	199.817
Acetone	CH_3COCH_3		7.02447	1161.0	224
Acetonitrile	CH_3CN		7.11988	1314.4	230
Acrylonitrile	C_3H_3N	−20 to +140	7.03855	1232.53	222.47
Ammonia	NH_3	−83 to +60	7.36050	926.132	240.17
Benzene	C_6H_6	+8 to +103	6.90565	1211.033	220.790
Benzoic acid	C_6H_5COOH	96 to 250	7.3533	1771.4	145.67
Bromine	Br_2		6.87780	1119.68	221.38
n-Butanol	C_4H_9OH	+15 to +131	7.47680	1362.39	178.77
Butadiene, 1,3	C_4H_6	−58 to +15	6.84999	930.546	238.854
Carbon disulfide	CS_2	3 to 80	6.94279	1169.11	241.59
Chlorine	Cl_2		6.93790	861.34	246.33
Chloroform	$CHCl_3$	−35 to 61	6.4934	929.44	196.03
Diethanolamine	$(C_2H_5O)_2NH$	194 to 241	8.1388	2327.9	174.4
Diethylamine	$(C_2H_5)_2NH$	31 to 61	5.8016	583.30	144.1
Dimethylamine	$(CH_3)_2NH$	−72 to +6.9	7.08212	960.242	221.67
Ethanol	C_2H_5OH	−2 to +100	8.04494	1554.3	222.65
Ethanolamine	C_2H_7ON	65 to 171	7.4568	1577.67	173.37
Ethyl acetate	$CH_3COOC_2H_5$	15 to 76	7.10179	1244.95	217.88
Ethylamine	$C_2H_5NH_2$	−20 to +90	7.05413	987.31	220.0
Ethylbenzene	C_8H_{10}	26 to 164	6.95719	1424.255	213.21
Ethylene glycol	$C_2H_6O_2$	50 to 200	8.0908	2088.9	203.5
Ethylene oxide	C_2H_4O	−49 to +12	7.12843	1054.54	237.76
Formic acid	CH_2O_2	37 to 101	7.5818	1699.2	260.7
Glycerol	$C_3H_8O_3$	183 to 260	6.165	1036	28
n-Heptane	C_7H_{16}	−2 to +124	6.89677	1264.90	216.54
n-Hexane	C_6H_{14}	−25 to 92	6.87601	1171.17	224.41
Hydrogen cyanide	HCN	−16 to 46	7.5282	1329.5	260.4
Hydrogen peroxide	H_2O_2		7.96917	1886.76	220.6
Isopentane	C_5H_{12}		6.78967	1020.012	233.097
Isopropanol	C_3H_7OH	0 to 100	8.11778	1580.92	219.61
Lactic acid	$C_3H_6O_3$		8.06	1823.7	134
Methanol	CH_3OH	−14 to 65	7.89750	1474.08	229.13
		65 to 110	7.97328	1515.14	232.85
Methyl acetate	CH_3COOCH_3	1 to 56	7.0652	1157.63	219.73
Methyl ethyl ketone	$CH_3COC_2H_5$		6.97	1210	216
Naphthalene (s)	$C_{10}H_8$	86 to 250	7.01065	1733.71	201.86
(l)		125 to 218	6.8181	1585.86	184.82
Nitrogen	N_2		6.49457	255.68	266.55
n-Octane	C_8H_{18}	19 to 152	6.91868	1351.99	209.15
Oxygen	O_2		6.69144	319.013	266.697
n-Pentane	C_5H_{12}	−50 to 58	6.85221	1064.63	233.01
Phosgene	$COCl_2$	−68 to 68	6.84297	941.25	230
Phenol	C_6H_5OH	107 to 182	7.133	1516.79	174.95
n-Propanol	C_3H_7OH	2 to 120	7.84767	1499.21	204.64
Propionic acid	C_2H_5COOH	56 to 139	6.403	950.2	130.3
Silicon tetrachloride	$SiCl_4$	0 to 53	6.85726	1138.92	228.88
Styrene	C_8H_8	32 to 82	7.14016	1574.51	224.09
Tetramethyl lead	$C_4H_{12}Pb$	0 to 60	6.9377	1335.3	219.1
Toluene	C_7H_8	6 to 137	6.95464	1344.8	219.48
Water	H_2O	0 to 60	8.10765	1750.286	235.0
		60 to 150	7.96681	1668.21	228.0
n-Xylene	C_8H_{10}	32 to 172	6.99891	1474.679	213.69
m-Xylene	C_8H_{10}	28 to 166	7.00908	1462.266	215.11
p-Xylene	C_8H_{10}	27 to 166	6.99052	1453.43	215.31

Source: *Lange's Handbook of Chemistry*, 14th ed and NIST Chemistry Webbook.

Table A.1

Specific Gravities of Selected Engineering Materials

(a) Common Manometer Liquids at 20°C

Liquid	Specific Gravity
E.V. Hill blue oil	0.797
Meriam red oil	0.827
Benzene	0.879
Dibutyl phthalate	1.04
Monochloronaphthalene	1.20
Carbon tetrachloride	1.595
Bromoethylbenzene (Meriam blue)	1.75
Tetrabromoethane	2.95
Mercury	13.55

Source: Data from References [1–3].

Table A.1

Specific Gravities of Selected Engineering Materials *(continued)*

(b) Common Materials

Material	Specific Gravity (—)
Aluminum	2.64
Balsa wood	0.14
Brass	8.55
Cast Iron	7.08
Concrete (cured)	2.4[a]
Concrete (liquid)	2.5[a]
Copper	8.91
Ice (0°C)	0.917
Lead	11.4
Oak	0.77
Steel	7.83
Styrofoam (1 pcf[b])	0.0160
Styrofoam (3 pcf)	0.0481
Uranium (depleted)	18.7
White pine	0.43

Source: Data from Reference [4].
[a]depending on aggregate.
[b]pounds per cubic foot.

Table A.2

Physical Properties of Common Liquids at 20°C

Liquid	Isentropic Bulk Modulus[a] (GN/m^2)	Specific Gravity (—)
Benzene	1.48	0.879
Carbon tetrachloride	1.36	1.595
Castor oil	2.11	0.969
Crude oil	—	0.82–0.92
Ethanol	—	0.789
Gasoline	—	0.72
Glycerin	4.59	1.26
Heptane	0.886	0.684
Kerosene	1.43	0.82
Lubricating oil	1.44	0.88
Methanol	—	0.796
Mercury	28.5	13.55
Octane	0.963	0.702
Seawater[b]	2.42	1.025
SAE 10W oil	—	0.92
Water	2.24	0.998

Table A.8

Properties of Water (SI Units)

Temperature, T (°C)	Density, ρ (kg/m^3)	Dynamic Viscosity, μ (N · s/m^2)	Kinematic Viscosity, ν (m^2/s)	Surface Tension, σ (N/m)	Vapor Pressure, p_v (kPa)	Bulk Modulus, E_v (GPa)
0	1000	1.76E-03	1.76E-06	0.0757	0.661	2.01
5	1000	1.51E-03	1.51E-06	0.0749	0.872	
10	1000	1.30E-03	1.30E-06	0.0742	1.23	
15	999	1.14E-03	1.14E-06	0.0735	1.71	
20	998	1.01E-03	1.01E-06	0.0727	2.34	2.21
25	997	8.93E-04	8.96E-07	0.0720	3.17	
30	996	8.00E-04	8.03E-07	0.0712	4.25	
35	994	7.21E-04	7.25E-07	0.0704	5.63	
40	992	6.53E-04	6.59E-07	0.0696	7.38	
45	990	5.95E-04	6.02E-07	0.0688	9.59	
50	988	5.46E-04	5.52E-07	0.0679	12.4	2.29
55	986	5.02E-04	5.09E-07	0.0671	15.8	
60	983	4.64E-04	4.72E-07	0.0662	19.9	
65	980	4.31E-04	4.40E-07	0.0654	25.0	
70	978	4.01E-04	4.10E-07	0.0645	31.2	
75	975	3.75E-04	3.85E-07	0.0636	38.6	
80	972	3.52E-04	3.62E-07	0.0627	47.4	
85	969	3.31E-04	3.41E-07	0.0618	57.8	
90	965	3.12E-04	3.23E-07	0.0608	70.1	2.12
95	962	2.95E-04	3.06E-07	0.0599	84.6	
100	958	2.79E-04	2.92E-07	0.0589	101	

THE END